

Chapter 9, Problem 10P

Problem

Design a problem to help other students better understand phasors.

Step-by-step solution

Step 1 of 3

Design a problem to help other students better understand phasors.

Although there are many ways to work this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

Given that $z_1 = 6 - j8$, $z_2 = 10 \angle -30^\circ$, and $z_3 = 8e^{-j120^\circ}$, find:

(a) $z_1 + z_2 + z_3$

Step 2 of 3

(b) $z_1 z_2 / z_3$

Solution

(a) $z_1 = 6 - j8$, $z_2 = 8.66 - j5$, and $z_3 = -4 - j6.9282$

$$z_1 + z_2 + z_3 = (10.66 - j19.928)\Omega$$

Step 3 of 3

(b) $\frac{z_1 z_2}{z_3} = [(10 \angle -53.13^\circ)(10 \angle -30^\circ) / (8 \angle -120^\circ)] = 12.5 \angle 36.87^\circ \Omega = (10 + j7.5) \Omega$